

Euclid, Book I, Proposition I.12

To Draw a Perpendicular to a Given Straight Line from a Point Outside It

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1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a given point lying outside that line.

Classically, Euclid obtains the perpendicular by choosing a point on the opposite side of the given line, drawing a circle centered at the external point, and using the two intersections of the circle with the given line. The segment joining these intersections is bisected by Proposition I.10, and Proposition I.8 is then applied to prove that the line joining the external point to the midpoint is perpendicular to the given line.

The reconstruction over the Hilbert foundation follows a different synthetic route. No circle-line intersection theorem is used. Instead, a second point is constructed on the opposite side of the given line so that two distinct points of the line are equidistant from the original external point and the constructed point.

This leads naturally to the perpendicular-bisector configuration. The required midpoint property is isolated in the general theorem

`hilbert_equidistant_line_bisects_segment`

which states that if two distinct points of a line are equidistant from two points lying on opposite sides of that line, then the intersection of the joining segment with the line bisects that segment.

A second auxiliary result,

`hilbert_collinear_angle_transport`

encapsulates the angle-order argument needed in the proof of this perpendicular-bisector theorem.

Consequently, Proposition I.12 provides another example of the role of the Book Zero layer: a comparatively elaborate geometric configuration is separated into reusable synthetic lemmas before the final Euclidean proposition is formulated.

2 The Classical Construction

Euclid's twelfth proposition asks for a perpendicular to a given straight line from a point outside it.

Let AB be the given straight line and let C be the given point outside it.

2.1 Final Configuration

The objective is to determine a point H on the straight line AB such that

$$CH \perp AB.$$

Euclid constructs two points E and G on AB satisfying

$$CE \cong CG,$$

and then bisects EG at H .

The final configuration therefore satisfies

$$E - H - G, \quad EH \cong HG, \quad CE \cong CG.$$

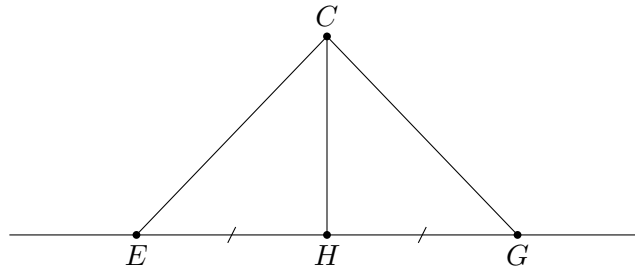


Figure 1: Final configuration of Euclid's construction for Proposition I.12.

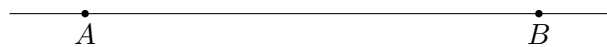
2.2 Construction Algorithm

The construction uses Proposition I.10 as an already established construction primitive.

2.2.1 Step 1. Given Line and External Point

Let AB be the given straight line and let C be a point not lying on AB .

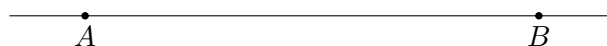
C
•



2.2.2 Step 2. Choose a Point on the Opposite Side

Choose a point D on the opposite side of the straight line AB from C .

C
•



•
 D

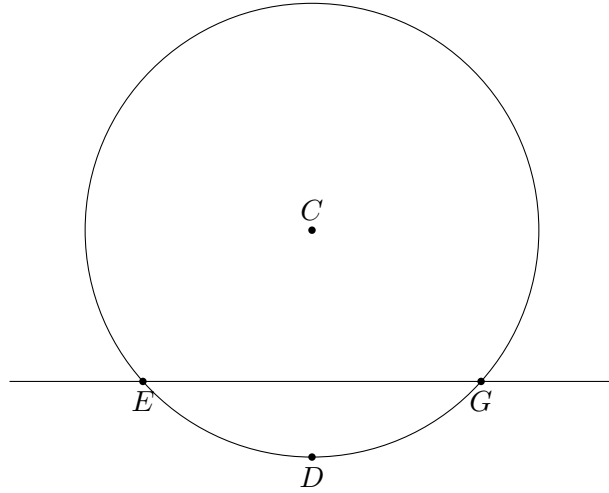
2.2.3 Step 3. Draw the Circle Centered at C

Draw the circle centered at C and passing through D .

Let the circle meet the given straight line at E and G .

Since E and G lie on the same circle centered at C ,

$$CE \cong CG.$$



2.2.4 Step 4. Bisect EG

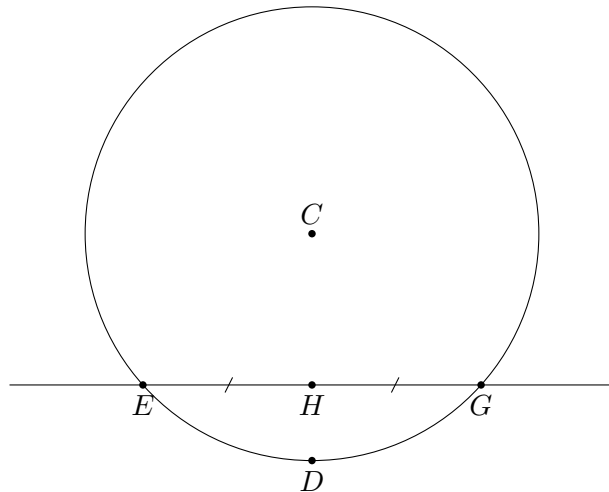
Apply Proposition I.10 to bisect the segment EG at H .

Thus

$$E - H - G$$

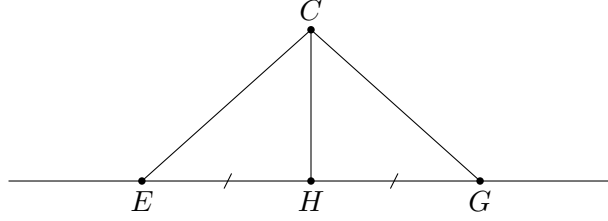
and

$$EH \cong HG.$$



2.2.5 Step 5. Join CH

Join the external point C to the midpoint H .



The two triangles EHC and GHC now satisfy

$$EH \cong HG, \quad HC \cong HC, \quad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are adjacent angles. By Definition 10 they are right angles. Therefore

$$CH \perp AB.$$

The required perpendicular has been constructed.

3 Proof

The Hilbert reconstruction does not reproduce Euclid's circle construction.

Let A and B be two distinct points on the given straight line b , and let P be the prescribed point outside b .

The first part of the construction produces a point Q on the opposite side of b from P such that

$$AP \cong AQ$$

and

$$BP \cong BQ.$$

Thus the two distinct points A and B of the line b are both equidistant from P and Q .

Since P and Q lie on opposite sides of b , the segment PQ meets b . Let H be the intersection point. Then

$$P - H - Q.$$

The central step is supplied by the Book Zero theorem

`hilbert_equidistant_line_bisects_segment`

Applied to the present configuration, it yields

$$PH \cong HQ.$$

Hence H is the midpoint of the segment PQ .

It remains to prove that the line through P and H is perpendicular to b .

Since $A \neq B$, at least one of the points A and B is distinct from H . Choose such a point and call it R .

Because R is either A or B , the construction already provides

$$RP \cong RQ.$$

Consider the triangles

$$\triangle RHP \quad \text{and} \quad \triangle RHQ.$$

They satisfy

$$RH \cong RH, \quad PH \cong HQ, \quad RP \cong RQ.$$

Hilbert SSS therefore gives

$$\angle RHP \cong \angle RHQ.$$

Now extend RH through H to a point C , so that

$$R - H - C.$$

Since

$$P - H - Q,$$

the two straight lines RC and PQ intersect at H . The vertical-angle theorem gives

$$\angle RHQ \cong \angle PHC.$$

Consequently,

$$\angle RHP \cong \angle PHC.$$

The angles RHP and PHC are adjacent angles, because $R - H - C$. By the definition of a right angle,

$$\text{HilbertRightAngle}(R, H, P).$$

Therefore the straight line through P and H is perpendicular to the given straight line b . This proves Proposition I.12.

4 Proof Architecture

The formal proof separates Proposition I.12 into two reusable Book Zero lemmas and one final construction theorem.

The first auxiliary result is

`hilbert_collinear_angle_transport`

It transports an angle congruence when one side of the angle is replaced by another point on the same supporting line. The proof handles the two possible ray configurations: direct transport along the same ray, and transport through supplementary angles when the new ray is opposite.

The second auxiliary result is

`hilbert_equidistant_line_bisects_segment`

Suppose two distinct points A and B lie on a line b , while P and Q lie on opposite sides of b , and

$$AP \cong AQ, \quad BP \cong BQ.$$

If the segment PQ meets b at H , then

$$PH \cong HQ.$$

The logical structure is therefore

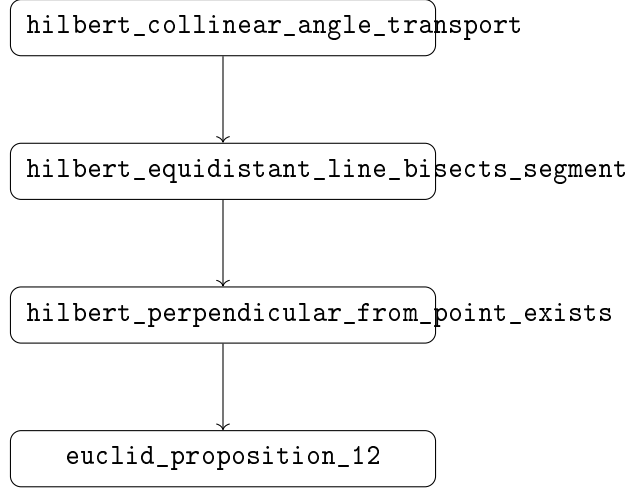


Figure 2: Logical structure of Proposition I.12 over the Book Zero layer.

The main construction theorem

`hilbert_perpendicular_from_point_exists`

uses the perpendicular-bisector result to obtain the midpoint H of PQ . It then chooses a point R on the given line distinct from H , applies Hilbert SSS to the triangles RHP and RHQ , and uses vertical angles together with the definition of a right angle.

Finally,

`euclid_proposition_12`

is a thin interface theorem exposing this general Hilbert result as Euclid's Proposition I.12.

5 Hilbert Reconstruction

The Hilbert reconstruction preserves the geometric objective of Proposition I.12 but does not reproduce Euclid's circle construction.

Let A and B be distinct points of the given line b , and let P be the prescribed point outside b .

The first task is to construct a point Q on the opposite side of b from P such that

$$AP \cong AQ, \quad BP \cong BQ.$$

To obtain Q , extend PA through A to a point S . Thus P and S lie on opposite sides of b .

Using Hilbert angle construction at A , construct a ray on the side of b containing S whose angle with AB is congruent to the angle formed by AP and AB . On this ray lay off a segment congruent to AP . Denote its endpoint by Q .

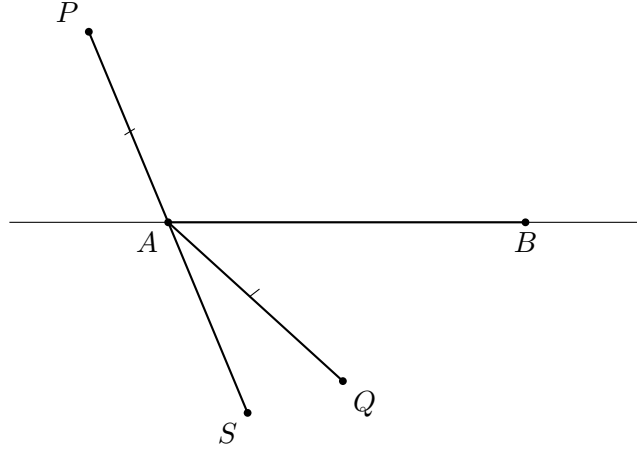


Figure 3: Construction of the point Q . The point S is obtained by extending PA through A . A ray is constructed on the side of the base containing S , and AP is laid off on this ray so that $AQ \cong AP$.

The construction gives

$$AP \cong AQ.$$

The two triangles

$$\triangle APB \quad \text{and} \quad \triangle AQB$$

now have

$$AP \cong AQ, \quad AB \cong AB,$$

and the included angles at A are congruent. By SAS,

$$BP \cong BQ.$$

Moreover, the construction places Q on the opposite side of b from P . Hence the segment PQ intersects b at a point H .

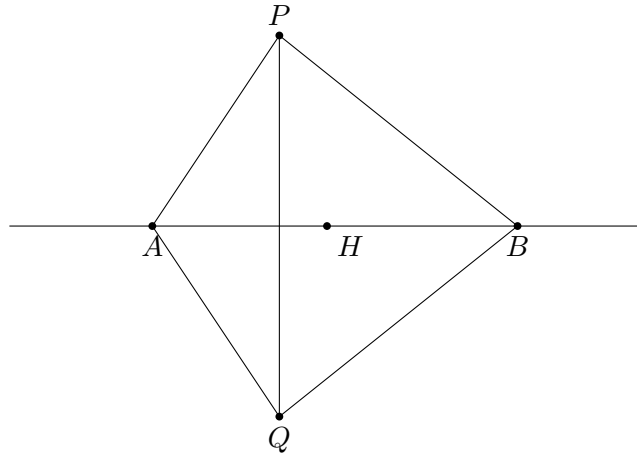


Figure 3: The perpendicular-bisector configuration used in the Hilbert reconstruction.

At this point the general Book Zero theorem

`hilbert_equidistant_line_bisects_segment`

applies. Since

$$AP \cong AQ, \quad BP \cong BQ,$$

and A and B are distinct points of b , the intersection point H bisects PQ :

$$PH \cong HQ.$$

Thus the difficult incidence and angle-order argument has been encapsulated before Proposition I.12 itself is completed.

To obtain the right angle, choose one of A and B distinct from H and call it R . Since R is one of the two equidistant points,

$$RP \cong RQ.$$

Together with

$$PH \cong HQ \quad \text{and} \quad RH \cong RH,$$

Hilbert SSS applied to

$$\triangle RHP \quad \text{and} \quad \triangle RHQ$$

gives congruent angles at H .

Since $P - H - Q$, these congruent angles determine a perpendicular to b at H . In the formal development this last step is expressed using vertical angles and the predicate

`HilbertRightAngle`

The reconstruction therefore replaces Euclid's use of a circle and Propositions I.10 and I.8 by a Hilbert construction based on angle construction, segment layoff, SAS, the perpendicular-bisector theorem, and SSS.

The resulting geometric object is nevertheless the same: a point H on the given line such that the line through P and H is perpendicular to the given line.

6 Lean Formalization

The Lean formalization separates the reusable geometric content from the final statement of Proposition I.12.

The main construction theorem is

```
theorem hilbert_perpendicular_from_point_exists
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hABase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P
```

Here H is the foot of the perpendicular and R is a second point of the given line used to determine its direction at H .

The presence of R in the conclusion is important. The intersection point H may coincide with either A or B , so neither of the original points can be fixed in advance as the first point in the right-angle predicate. Since

$$A \neq B,$$

at least one of them is distinct from H and can therefore be chosen as R .

The formal proof first constructs a point Q on the opposite side of the given line from P . The construction establishes

$$AP \cong AQ$$

and, by SAS,

$$BP \cong BQ.$$

The opposite-side relation supplies an intersection point H with

$$P - H - Q.$$

The reusable Book Zero theorem

`hilbert_equidistant_line_bisects_segment`

then gives

$$PH \cong HQ.$$

The remaining argument selects R , applies Hilbert SSS to the triangles RHP and RHQ , and converts the resulting angle congruence into a right angle by means of vertical angles.

The Euclidean proposition itself is exposed by the wrapper

```
theorem euclid_proposition_12
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P := by

exact
hilbert_perpendicular_from_point_exists
Geo
A B P
base
hAB
hAbase
hBbase
hPbase
```

Thus the theorem carrying Euclid's name is deliberately only a thin wrapper. The geometric work is performed by the Hilbert construction theorem and by the reusable results already established in the Book Zero layer.

7 Discussion

Proposition I.12 reveals a substantial difference between Euclid’s constructional language and the Hilbert/Book Zero architecture.

Euclid obtains the perpendicular through a concrete circle construction. The circle centered at the external point produces two points of the given line at equal distance from that point. Proposition I.10 supplies their midpoint, and Proposition I.8 converts the resulting three side equalities into equality of the adjacent angles at the midpoint.

The Hilbert reconstruction does not attempt to reproduce this sequence literally. In particular, no circle is introduced. Instead, the essential geometric configuration behind Euclid’s construction is isolated:

$$AP \cong AQ, \quad BP \cong BQ.$$

Thus two distinct points A and B of the given line are equidistant from P and Q . The line through A and B therefore plays the role of the perpendicular-bisector axis of PQ .

This observation leads to the theorem

`hilbert_equidistant_line_bisects_segment`

which extracts the reusable geometric content of the configuration. Once the segment PQ intersects the line at H , the theorem gives

$$PH \cong HQ.$$

The proposition is therefore no longer organized around a particular circle construction. It is organized around an abstract property of two equidistant points and the line determined by them.

The proof of the perpendicular-bisector theorem itself exposed another reusable operation. When an angle is known relative to one point of a line, replacing that point by another collinear point requires attention to the order of the points: the relevant ray may remain unchanged or become the opposite ray. This argument is encapsulated in

`hilbert_collinear_angle_transport`

rather than being repeated inside Proposition I.12.

This is characteristic of the role of Book Zero. Elementary incidence, order, congruence, ray, and angle manipulations are packaged as synthetic lemmas. Later Euclidean propositions can then operate with larger geometric units instead of repeatedly descending to the Hilbert axioms.

There is also an important difference from the preceding elementary propositions. Proposition I.12 does not collapse to a single previously available Book Zero operation. Its reconstruction required the identification of genuinely reusable intermediate geometry. In this sense, the formalization of I.12 contributed new infrastructure to Book Zero rather than merely consuming existing infrastructure.

The final theorem

`euclid_proposition_12`

is consequently a wrapper around

`hilbert_perpendicular_from_point_exists`

The wrapper records the Euclidean proposition in the formal library, while the underlying theorem records the more general Hilbert construction on which it depends.

Proposition I.12 therefore illustrates the intended division of labor in the development:

Hilbert axioms	→	elementary incidence, order, and congruence theory
Book Zero	→	reusable synthetic geometric mechanisms
Hilbert construction theorem	→	existence of the required perpendicular
Euclid wrapper	→	Proposition I.12

The result is not a literal transcription of Euclid's proof. It is a reconstruction of the same geometric theorem over the Hilbert foundation, with the intermediate reasoning reorganized into reusable components.